

7. الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس). *

UJWJ

 **Will be reviewed**

8. Which type of process would benefit most from a high-performance CPU? *

- ☒ CPU-bound process
- ☐ Neither
- ☐ Both equally
- ☐ I/O-bound process

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9. In non-preemptive scheduling, a running process: *

- ☐ Can be interrupted by the OS
- ☐ Is always interrupted after a fixed time quantum
- ☒ Cannot be interrupted until it completes or voluntarily releases the CPU
- ☐ Is always interrupted when a higher priority process arrives

 **Will be reviewed**

10. A race condition occurs when: *

- ☐ A process is starved of CPU time.
- ☒ Two or more processes access a shared resource concurrently.
- ☐ A process is waiting for a resource that is held by another process.

- ☐ A process is terminated abruptly.

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11. Below the structure of the Readers & Writers problem is shown using the semaphores mutex & wrt and an integer variable readCount.

READER

```
wait(mutex);
readCount++;
if (readCount == 1)
    wait(wrt);
signal(mutex);
... reading section...
wait(mutex);
readCount--;
if (readCount == 0)
    signal(wrt);
signal(mutex);
```

WRITER

```
wait(wrt);
.....writing section.....
signal(wrt);
```

In the above algorithm what are the initial values of mutex, wrt, and readCount? *

- ☒ mutex = 1, wrt = 1, readCount = 0
- ☐ mutex = 1, wrt = 1, readCount = 1
- ☐ mutex = 0, wrt = 1, readCount = 0
- ☐ mutex = 1, wrt = 0, readCount = 0

 Will be reviewed

12. A disadvantage of the First-Come-First-Served (FCFS) scheduling algorithm is: *

- ☐ It can lead to starvation
- ☒ It is not fair

- ☐ It is difficult to implement
- ☐ It can lead to long wait times for short processes

 **Will be reviewed**

13. A disadvantage of the Shortest Job First (SJF) scheduling algorithm is: *

- ☐ It is difficult to implement
- ☐ It is not fair
- ☒ It can lead to starvation of long processes
- ☐ It can lead to long wait times for short processes

 **Will be reviewed**

14. Peterson's solution is limited to: *

- ☐ None of the above
- ☐ A fixed number of processes
- ☐ Two processes
- ☒ Any number of processes

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15. What is the role of the interrupt vector table in handling interrupts? *

- ☐ Defines memory access permissions
- ☒ Maps interrupt types to their corresponding interrupt service routine (ISR) locations
- ☐ Controls the speed of the CPU

- ☐ Stores the location of all programs in memory

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16. What is the main constraint in the Readers-Writers Problem? *

- ☐ Multiple writers can access the resource simultaneously.
- ☐ One reader and one writer can access the resource simultaneously.
- ☐ Only one process can access the resource at any time.
- ☒ Multiple readers can access the resource simultaneously, but writers must have exclusive access.

 **Will be reviewed**

17. What is the main purpose of Direct Memory Access (DMA)? *

- ☐ To manage memory allocation for user programs
- ☐ To provide a user interface for interacting with the system
- ☒ To transfer data between I/O devices and memory without CPU intervention
- ☐ To schedule processes for execution

 **Will be reviewed**

18. The primary function of the Bootstrap program is to: *

- ☐ Connect to the internet
- ☐ Run games and multimedia
- ☐ Manage user applications

- ☒ Initialize the system and load the operating system

 **Will be reviewed**

19. The Producer-Consumer Problem involves: *

- ☐ Multiple producers and a single consumer
- ☐ A single producer and multiple consumers
- ☐ Multiple producers and multiple consumers
- ☒ A single producer and a single consumer

 **Will be reviewed**

20. Peterson's solution guarantees: *

- ☐ Mutual exclusion
- ☒ All of the above
- ☐ Progress
- ☐ Bounded waiting

 **Will be reviewed**

21. Suppose that a process is in “Blocked” state waiting for some I/O service. When the service is completed, it goes to the _____ *

- ☐ Terminated
- ☒ Ready state
- ☐ Suspended state

☐ Running state

 **Will be reviewed**

22. A test-and-set instruction is an atomic operation that: *

- ☐ Reads a value from memory
- ☐ None of the above
- ☐ Sets a value in memory
- ☒ Reads a value from memory, sets it to a new value, and returns the old value

 **Will be reviewed**

23. The real difficulty with SJF in short term scheduling is _____ *

- ☒ knowing the length of the next CPU request
- ☐ none of the mentioned
- ☐ it is too good an algorithm
- ☐ it is too complex to understand

 **Will be reviewed**

24. What is the primary purpose of the Test and Set instruction? *

- ☒ To provide a hardware-based solution for mutual exclusion.
- ☐ To schedule processes.
- ☐ To manage memory allocation.

☐ To handle I/O requests.

 **Will be reviewed**

25. Assume three jobs arrive at approximately the same time, but Job A arrives slightly before Job B, and Job B arrives slightly before job C. Job A requires 400 ms of CPU, Job B is 800 ms, and Job C is 1200 ms. Assume a time-slice of 600 msec. Given a RR scheduler, what is the turnaround time of job B? *

☐ 2400 ms

☐ 1600 ms

☐ 800 ms

☒ 1800 ms

☐ 1200 ms

 **Will be reviewed**

26. Assume three jobs arrive at approximately the same time, but Job A arrives slightly before Job B, and Job B arrives slightly before job C. Job A requires 400 ms of CPU, Job B is 800 ms, and Job C is 1200 ms. Assume a time-slice of 600 msec. Given a RR scheduler, what is the waiting time of job C? *

☐ 2400 ms

☐ 800 ms

☒ 1200 ms

☐ 400 ms

 **Will be reviewed**

27. Consider the following set of processes, the length of the CPU burst time given in milliseconds. The processes are assumed to have arrived all at time 0, at order P1, P2, P3,

P4.Process	Burst time
P1	6
P2	8
P3	7
P4	3

Assuming the above process being scheduled with the SJF scheduling algorithm.

*

- ☒ The waiting time for process P1 is 3ms
- ☐ The waiting time for process P1 is 16ms
- ☐ The waiting time for process P1 is 0ms
- ☐ The waiting time for process P1 is 9ms



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